

Keshav Mittal

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SUMMARY

Biomedical Engineering postgraduate at UNSW with hands-on experience in prosthetic control, wearable biomedical sensing, rehabilitation technology, and assistive device prototyping. Skilled in MATLAB, COMSOL, OpenSim, Python, biomechanical simulation, and experimental research. Passionate about developing practical medical devices and human-centred technologies for prosthetics, mobility, and healthcare innovation.

EDUCATION

University of New South Wales (UNSW) Sydney Feb 2025 – May 2027
Master of Biomedical Engineering Sydney, Australia

- Current postgraduate student

Dr. D.Y. Patil Vidyapeeth, Pune Aug 2019 – Jun 2023
Bachelor of Technology in Biotechnology Pune, India

- CGPA: 9.29/10, First Class with Distinction

RELEVANT PROJECTS

- **UniSense – Dual-Mode Prosthetic Socket Mapping System:** Developing a wearable biomedical system for prosthetic socket optimization by combining EMG-based control mapping with pressure and temperature-based comfort mapping. Built around sensor integration, MATLAB-based analysis, and COMSOL-informed design support for improving prosthetic fit and control-site selection.
- **Simulation and Design of Clutch Mechanism with Passive Cable Routing:** Designed and simulated a passive clutch and cable-routing mechanism intended to support posture alteration during walking. Focused on mechanical design, movement assistance, cable path optimization, and practical translation into wearable rehabilitation or assistive systems.
- **Markerless Optical Myography for Bionic Control System:** Worked on developing a markerless, non-contact optical myography approach for prosthetic and bionic control applications. Contributed to machine learning strategy development, prototyping, experimentation, and model evaluation, achieving approximately 95% accuracy in control-related classification tasks.
- **UnMute – Auslan-to-Voice Assistive Communication System:** Ongoing project focused on developing a wearable sign-to-speech system using multiple IMU-based rings and hand-mounted sensors. The system aims to capture hand motion, classify sign gestures using machine learning, and convert recognized gestures into real-time voice output for accessible communication.
- **SPATIAL Coordination Operating System:** Ongoing concept exploring how spatial positioning, human movement, and task coordination can be captured and structured into an intelligent operating layer. The project aims to investigate spatial awareness, coordination logic, and real-time decision support for complex human-device or clinical environments.
- **TremoFlex – Tremor-Reducing Wearable Assistive Glove:** Designed a wearable assistive device concept aimed at reducing hand tremors using a tuned mass damper-based stabilization mechanism. Focused on mechanical design, user-centred rehabilitation needs, and improving fine motor control for people with essential tremor or osteoarthritis during daily tasks.

PROFESSIONAL EXPERIENCE

Indian Institute of Technology Delhi Jul 2023 – May 2024
Project Intern – RISE Lab Delhi, India

- Worked on biomedical research related to prosthetic control, optical myography, and assistive technology applications.
- Contributed to research activities involving literature review, experimental documentation, data interpretation, and technical communication.

Indian Institute of Technology Gandhinagar Jan 2023 – May 2023
Research Intern – Human Centered Robotics Lab Gandhinagar, India

- Supported research activities in human-centered robotics with focus on assistive and biomedical engineering applications.

- Gained exposure to robotics research workflows, experimental planning, technical documentation, and interdisciplinary collaboration.

Bioclinica Biosciences
Scientific Content Writer

Oct 2021 – Nov 2021
Visakhapatnam, India

- Developed scientific and biomedical content by translating complex biotechnology and healthcare concepts into clear written material.

LifegenBio
Marketing and Technical Intern

Sep 2020 – Dec 2020
Remote / India

- Supported technical and marketing activities related to biotechnology, bioinformatics, and scientific communication.

TECHNICAL SKILLS

- **Engineering and Simulation:** MATLAB, COMSOL Multiphysics, OpenSim, AutoCAD.
- **Programming and Data Tools:** Python, Biopython, basic C/C++, data analysis, signal processing fundamentals.
- **Biomedical and Hardware Skills:** EMG signal acquisition, wearable sensors, ESP32 prototyping, FSR sensors, thermistor sensing, biomedical instrumentation basics.
- **Design and Research Skills:** Medical device prototyping, biomechanics, prosthetics and rehabilitation technology, literature review, technical writing, MS Office.

AWARDS AND ACHIEVEMENTS

- Presented “*UniSense: A dual mode prosthetic socket for comfort mapping and control mapping*” at the Australasian Bioelectronics, Neurosensing and Neuromodulation Symposium 2026.
- Secured first position in speech competition during Swachhta Pakhwada at DYPBBI.
- Won silver medal and second position in Kabaddi during the annual function Cytosol 2020 at DYPBBI.
- Participated in a Science Innovation Competition presenting a working vertical tube gel electrophoresis method for extrapolating metabolite biomarkers for early cancer detection.

RESEARCH PUBLICATION

- Vivek Raj, **Keshav Mittal**, Manikandan Shenbagam, Jishnu Perunthattil, Srikumar Venkataraman, Biswarup Mukherjee. “*Towards Markerless, Non-Contact Optical Myography for Prosthetic Control of Multiple Degrees-of-Freedom.*” 2024.

LEADERSHIP AND EXTRA-CURRICULAR ACTIVITIES

- Served as General Secretary of the Student Council at Dr. D.Y. Patil Biotechnology and Bioinformatics Institute from Nov 2021 to Nov 2022.
- Served as Sports Secretary of the University Student Council at Dr. D.Y. Patil Vidyapeeth from Nov 2021 to Nov 2022.
- Served as Academic Secretary and Class Representative for B.Tech. Biotechnology, supporting student coordination and academic communication.
- Volunteered with the National Social Service program in social initiatives including blood donation, cleanliness drives, and tree plantation.

REFEREES

- **Dr. Biswarup Mukherjee**
Centre for Biomedical Engineering, Indian Institute of Technology Delhi
Project Guide